

Module 3

Configuring Interfaces and Switches

Learning Objectives

- Deploy networking devices.
- Explain network interfaces.
- Deploy common Ethernet switching features.
- Troubleshoot transceiver switching issues.

Lesson 3.1

Network Interfaces

Think About It: Network Interfaces

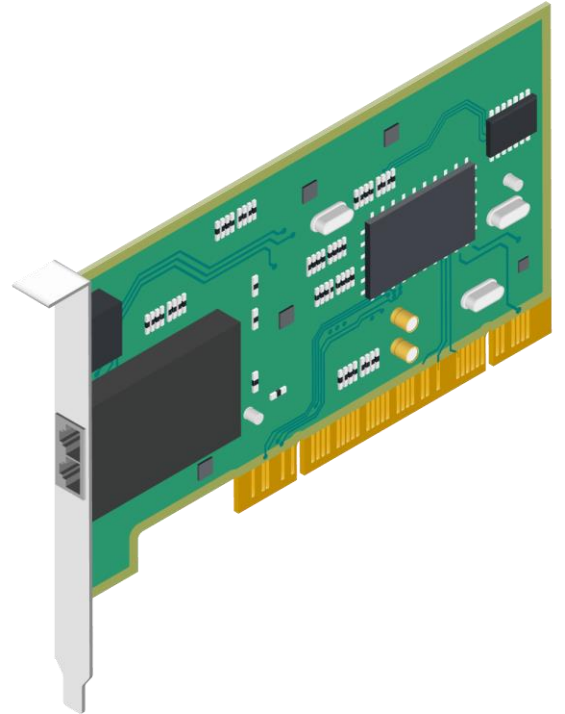
- What is a network interface?



Network Interface Cards

A network interface card (NIC):

- Transceiver component
- Connects the host to a transmission medium (wired or wireless)
- Can have multiple ports on same card
- Has a unique MAC address
- Operates at the Data Link layer



Symptoms of NIC Issues

Common symptoms of NIC issues include:

- Network connectivity failure
- Slow network speeds
- Intermittent connectivity
- Device manager error messages
- Network setting error messages



Modular Transceivers

Modular transceivers

- Terminate multiple types of cable and connector types
- Operate at the Data Link layer



Symptoms of Transceiver Issues

Mismatched ports

- No link
- Intermittent connection loss

Signal strength

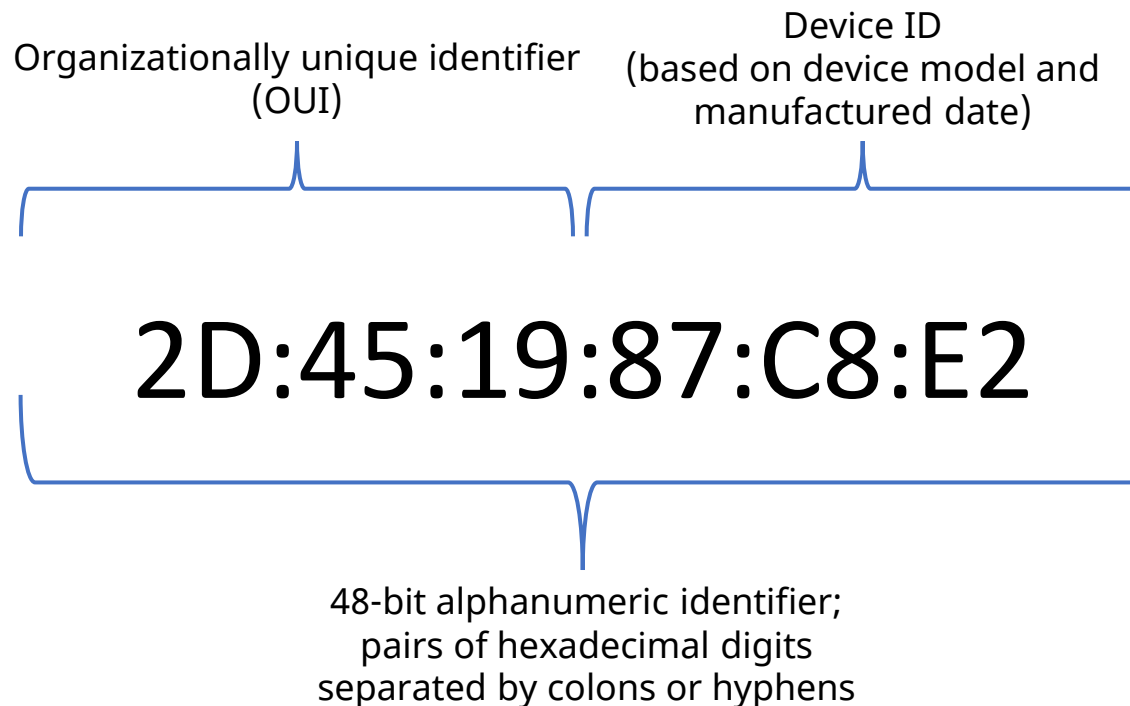
- Intermittent connections
- Packet loss
- Poor network performance



Ethernet Frame Format

Preamble Synchronization sequence	SFD Signals the start of the frame	Destination MAC Address of the recipient device	Source MAC Address of the sending device	Ether Type Protocol of the payload	Payload Data being transmitted	FCS Error-checking code
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MAC Address Format



MAC Broadcast Address

Preamble	SFD	Destination MAC	Source MAC	Ether Type	Payload	FCS
Synchronization sequence	Signals the start of the frame	Address of the recipient device	Address of the sending device	Protocol of the payload	Data being transmitted	Error-checking code



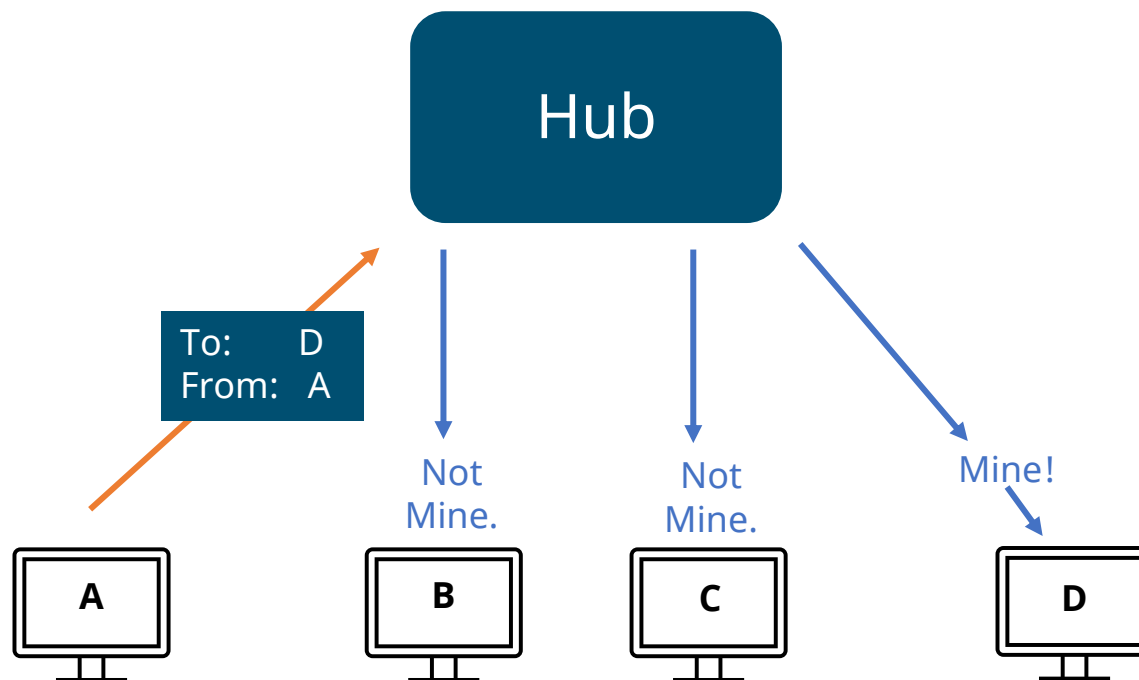
Broadcast Address
11:11:11:11:11:11

If MAC address is all 1s, all hosts on that network will receive and process the packet

Lesson 3.2

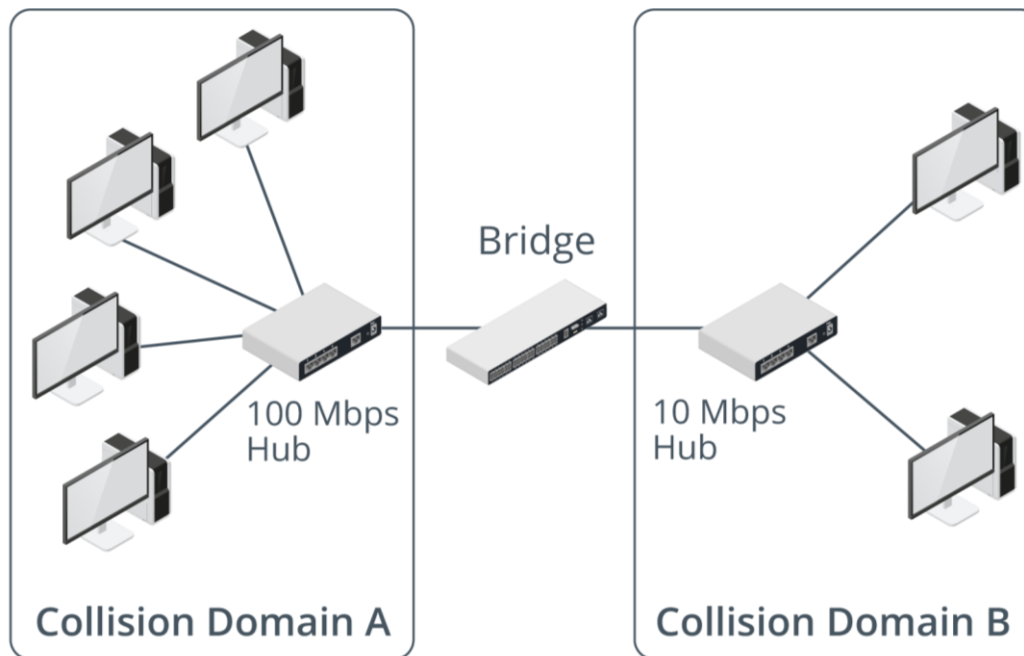
Ethernet Switches

Hubs



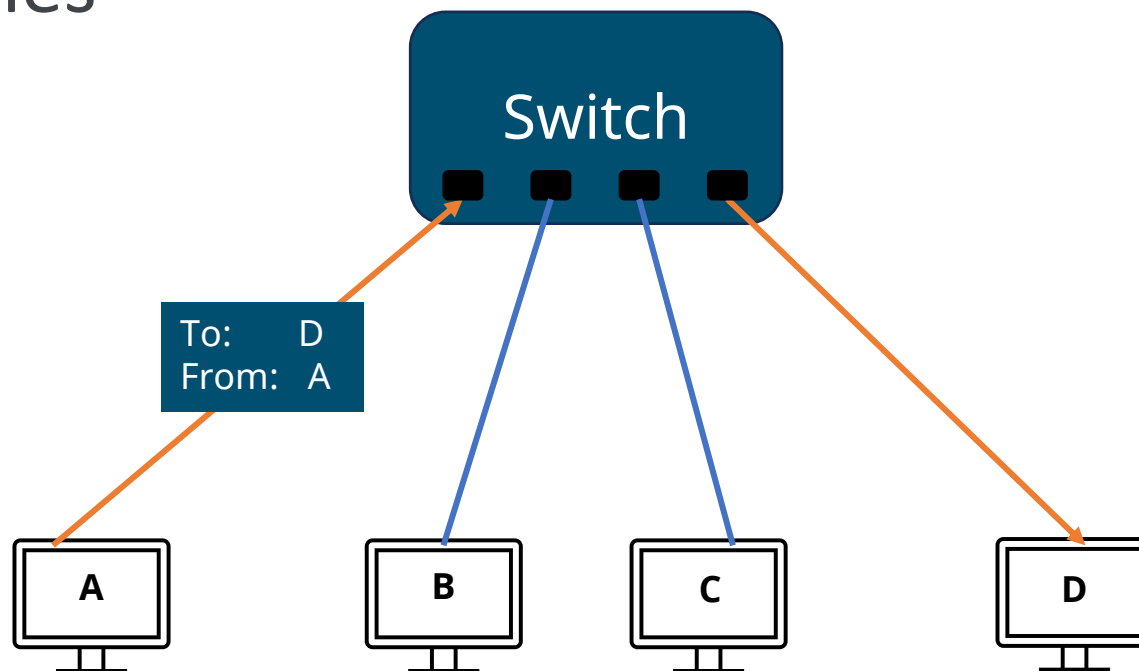
Hubs send transmissions from one port to every other port.

Bridges



Bridges separate physical network segments while keeping all nodes in the same logical network.

Switches



A switch sends transmissions from one port only to the destination port.

Ethernet Switch Types

Unmanaged vs. Managed

Managed switches can be configured

Unmanaged switches have no configuration options

Modular vs. Fixed

Modular can be configured with different numbers and types of ports

Fixed come with a set number of ports

Desktop vs. Rack

Desktop switches are free standing

Rack-mounted switches are designed fit into networking racks

Stackable

Can be connected together

Can be managed as a single unit

Activity: Matching



Improves network efficiency by dividing it into smaller, manageable sections

Activity: Matching



Hub

Switch

Bridge

Operates by creating a single shared communication path between multiple devices in a network

Activity: Matching



Intelligently directs data to a specific port based on MAC addresses, reducing network congestion

Activity: Matching



Hub

Switch

Bridge

Operates primarily at the physical layer

Activity: Matching



Learns the MAC addresses of devices connected to it and uses this information to forward data only to the intended recipient

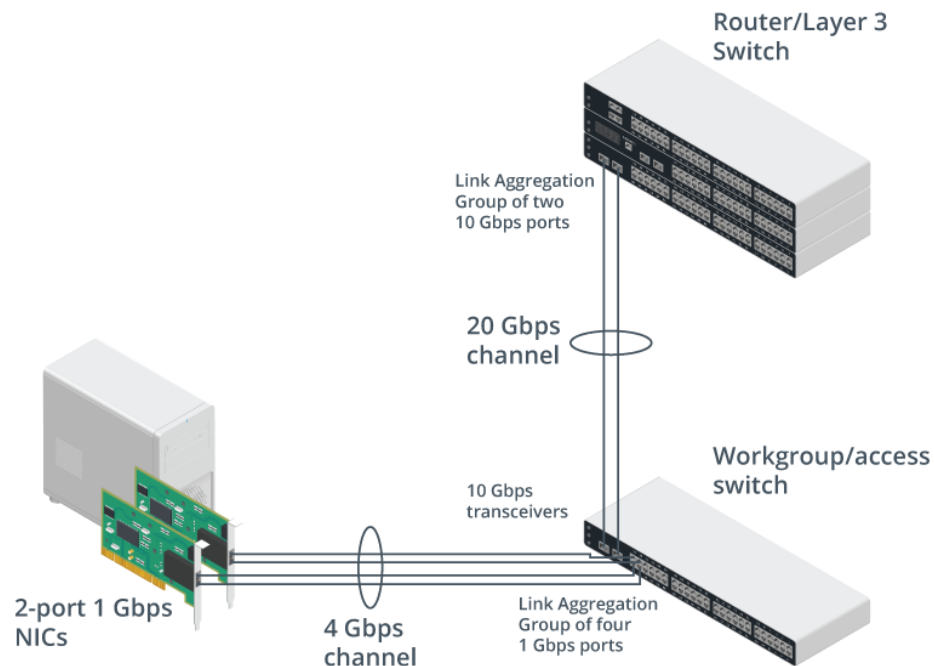
Lesson 3.3

Switch Port Configuration

Link Aggregation/NIC Teaming

Link aggregation

- Combining 2+ separate cabled links into a single logical channel
- Provides redundancy
- Cost effective



Maximum Transmission Unit

Standard ethernet frame

- Maximum transmission unit (MTU) is 1,500 bytes

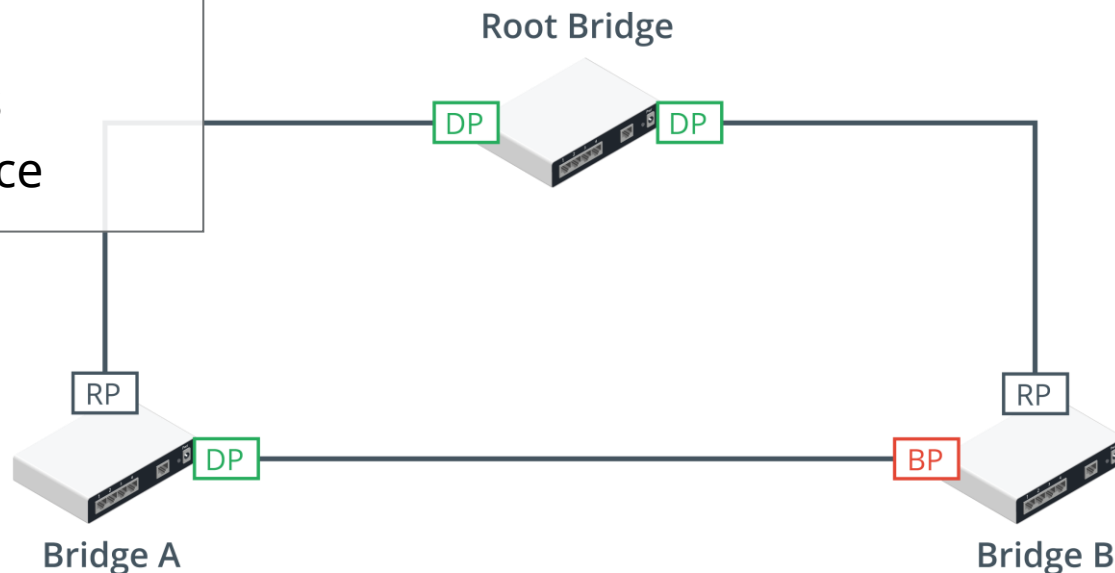
Jumbo frame

- Supports payload up to 9.216 bytes
- Reduces the number of frames transmitted
- Limited because they break Ethernet standards

Spanning Tree Protocol

Spanning Tree Protocol (STP)

- Layer 2 protocol
- Prevents bridge loops
- Provides fault tolerance



Power Over Ethernet

Power over Ethernet (PoE)

- Allows 1 cable to transmit both data and power to networked devices
- Simplifies network installation and expansion



Lesson 3.4

Switch Troubleshooting

Hardware Failure Issues

Hardware Failure Issue	Troubleshooting and Mitigation Steps
Power issues	• Verify stable power supply • Install UPS and secondary power sources
Network adapters	• Check for damaged ports or connectors • Test with alternative adapter if possible • Update or reinstall drivers
Switches/routers/modems	• Visually inspect for damage • Inspect indicator lights • Verify power supply and cabling connections • Restart the device following proper protocols
Overheating	• Check for proper ventilation around the device • Clean dust from intake or exhaust vents • Ensure cooling systems are operational

Port Status Indicators

	Solid green	The link is connected, but there is no traffic
	Flickering green	The link is operating normally (with traffic). The blink rate indicates the link speed
	No light	The link is not working, or the port is shut down
	Solid amber	The port is blocked by the spanning tree algorithm
	Blinking amber	A fault has been detected

Switch Show Commands

Show config

- Device's current configuration

Show startup-config

- Configuration device will use upon the next restart

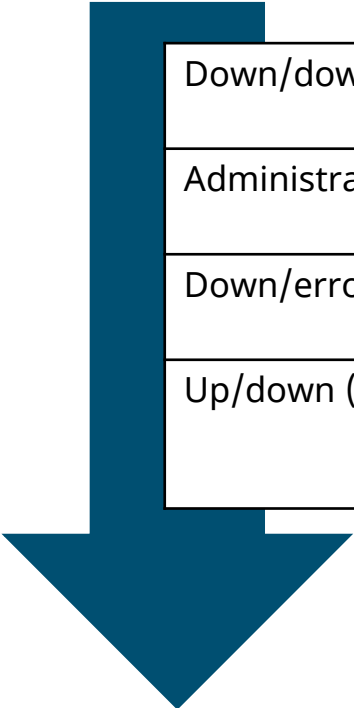
Show running-config

- Active configuration currently used by the device

Show interface

- Detailed information about the device's network interfaces

Switch Show Command Status



Down/down	Both the layer 1 (physical) and layer 2 (data link) connections are inactive.
Administratively down/down	The interface has been manually disabled by an administrator using the shutdown command
Down/error disabled	The interface has been automatically disabled due to a network error or policy violation
Up/down (suspended)	The physical layer is operational, but the data link layer is inactive due to administrative settings or errors

Interface Error Counters

Link state

- Checks if interface is up or down
- Immediate alert for downtime

Resets

- Count of manual and automatic restarts
- High frequency of resets should be monitored

Discards/drops

- Causes by checksum errors, mismatched MTUs, size anomalies, high load, ACL or VLAN configuration errors
- Used for troubleshooting



Common Interface Errors

Cyclic Redundancy Check (CRC) Errors

Frame's calculated checksum does not match the transmitted checksum

Indicative of noise, interference, or equipment malfunctions

Runt Frame Errors

Frames are smaller than the minimum frame size.

Results from collisions or damaged hardware.

Giant Frame Errors

Frames exceed the maximum allowed size.

Caused by misconfiguration or malfunctioning network devices

Network Loop and Broadcast Storm

Network Loops: Causes

- Redundant connections
- End-device side

Network Loops: Solutions

- Proper redundant connection configuration
- Implement spanning tree protocol (STP)
- Educate users

Broadcast Storms: Causes

- DHCP issues
- Very large broadcast domains

Broadcast Storms: Solutions

- Monitor DHCP traffic
- Segment the network

Power Over Ethernet Issues

Cabling Considerations

- Must use Cat 5e or better, recommend Cat 6A
- Importance of conductor thickness (23 AWG > 24 AWG for better performance)
- Shielded cabling helps disperse heat efficiently

Standard Compatibility

- Devices must negotiate the correct PoE mode (Alternative A, B, or Four-pair)
- Challenges with devices only supporting the first PoE standard

Power Budget Management

- Switches have a limited power budget (300–400W)
- Use "show power inline" command for monitoring

Summary

- **NIC vs. Transceiver:** NIC connects host to network (cable), transceiver adapts cable types
- **Hub vs. Switch:** Hub broadcasts to all devices, switch directs traffic to specific devices; bridge connects networks, keeps them logically unified
- **STP:** Prevents bridge loops and ensures network redundancy (Layer 2 protocol)